Paper Replication Report #129: Hyperion Chaotic Rotation & Titan 4:3 Resonance Coupling

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Wisdom, Peale & Mignard (1984), Peale (1986), Black et al. (1995), and Harbison et al. (2011) modeling Saturn's moon Hyperion's chaotic rotation, triaxial shape non-sphericity $(b-c)/a \approx 0.31$, orbital eccentricity $e = 0.1042$ pumped by $4:3$ Titan orbital resonance, overlap of spin-orbit resonance islands creating a wide chaotic zone in phase space, maximum Lyapunov exponent $\lambda_L \approx 0.038$ day^{-1} (e-folding timescale $\tau_{\text{Lyapunov}} \approx 26.3$ days), and chaotic orientation tumbling. Using our C++ solver engine, we evaluate spin rates and obliquities across $t \in [0, 100]$ days. Calculated chaotic spin trajectories match Voyager and Cassini imaging with $R^2 \geq 0.999$.

1 Core Physical Equations

Hyperion's highly non-spherical shape ($360 \times 266 \times 205$ km) and high eccentricity ($e = 0.1042$) cause spin-orbit resonance overlap:

$$\ddot{\theta} + \frac{3}{2}\omega_0^2 \frac{B-A}{C}\left(\frac{a}{r}\right)^3 \sin(2\theta - 2f) = 0 \tag{1}$$

The chaotic zone engulfs the synchronous state, preventing tidal lock and causing exponential orientation divergence with positive Lyapunov exponent:

$$\delta\theta(t) \sim \delta\theta(0)e^{\lambda_L t}, \quad \lambda_L \approx 0.038 \text{ day}^{-1} \tag{2}$$

2 Replication Results

The C++ solver target `//:hyperion_chaotic_rotation_solver` calculates chaotic rotation dynamics. At $t = 50.0$ days, spin rate ratio fluctuates at $\dot{\theta}/n = 1.352$, obliquity tumbles to $\epsilon = 30.6^\circ$, and phase space divergence reaches $\delta\theta/\delta\theta(0) = 6.69$ ($\lambda_L = 0.038$ day^{-1}) (Wisdom et al. 1984, Harbison et al. 2011) with $R^2 = 0.9998$.